

In accordance with the NHTSA’s mission to deregulate motor vehicle safety regulations and usher in an age of HAV prosperity, the agency ought to review – and perhaps retire – road-based standards within what is currently a primarily vehicle-oriented regulatory framework. While car-centric infrastructure initially supported the widespread adoption of personal vehicles – and the ability for individuals to operate them – this approach has since created lasting hindrances to the NHTSA’s ability to establish a tech-neutral performance standard. Road-based standards are a prime target for the agency’s specific goal to sunset “technical requirements that hinder the transition to technology-neutral, performance-based standards” (National Highway Traffic Safety Administration, 2026).

A tech-neutral performance standard, like those established in more agile infrastructure arrangements globally – e.g., China and the Netherlands – need not take the form of a single statute. In some cases, a de facto standard may emerge through exceptions and negative incentives rather than direct regulation. In the context of U.S. domestic policy, the current “gas guzzler” tax exemplifies direct regulation, while EV credits from the Inflation Reduction Act demonstrate how exception-based incentive structures can achieve similar ends. Exceptions and negative incentives align with the NHTSA’s goal of thorough deregulation and with domestic objectives, and could position the United States to adopt globally successful models of civil infrastructure and design. The NHTSA does not have the power to tax; it does, however, have the ability to establish a framework that can be legislated within.

All practical driving tests currently take place on roads. Since HAVs can operate in separate lanes from human drivers, road-based manufacturing standards have become inhibitory. Driving tests – including licensing exams – have not kept pace with modern HAV technology. Following the principle that modern problems require modern solutions, uniform testing

infrastructure is no longer sufficient for assessing a vehicle's safety. A more effective testing infrastructure, for both drivers and driverless vehicles, ought to assess performance and adaptability rather than compatibility with car-centric infrastructure.

Section 14.9.3.12.5 mandates adjacent test lanes, road uniformity, and solid-paved surfaces. Furthermore, a test road surface must be either concrete or asphalt and may not be bright white. While seemingly innocuous, these conditions lead to a final restriction: a road surface “may have pavement markings but shall be free of retroreflective material or elements that affect the outcome of the test” (FMVSS, 2004). These regulations hinder local municipalities' ability to update transportation infrastructure, innovate in road design, and pursue creative or regionally viable civil engineering efforts. Despite the United States' substantial capacity to build infrastructure, efforts to expand the interstate highway system rather than pursue decentralized innovation have hindered its ability to modernize infrastructure projects. It is difficult to reconcile the nation's status as the wealthiest in the world with its “C” rating from the American Society of Civil Engineers (ASCE, 2023, 2025). As such, attempts to integrate highly autonomous vehicle infrastructure – including dedicated lanes, autonomous buses, and other unique indicators – have been stymied by the enduring legacy of uniform, car-centric city planning. If vehicles are tested exclusively on conditions that mirror public roads, there is no metric to ensure their safety in novel environments.

Autonomous vehicles, in some cases, rely on lane markings and peripheral systems to assess environmental conditions. Uniform road infrastructure once effectively simulated the conditions of an average street or highway. For human drivers, occasional reminders suffice; autonomous vehicles, however, do not share this capacity for consciousness. Their lack of consciousness may instead justify shifting the burden of adaptability to the vehicle rather than

the infrastructure itself. Accordingly, it is imperative that the NHTSA remove inapplicable restrictions – such as those in S14.9.3.12.5 – to incentivize creative civil engineering and city planning. The incorporation of highways into cities need not remain a fact of life; in fact, it ought to depend on community values rather than federal mandate.

Globally, increased civic participation and local responsibility in urban design have yielded significant success. China’s recent investments in streamlined public transport – such as autonomous bus rapid transit systems in cities like Guangzhou – have been developed under performance-based standards that encourage innovation (United Nations Framework Convention on Climate Change, n.d.). These tech-neutral approaches have effectively supplemented regional civil engineering projects and corporate efforts to integrate HAVs into the transportation ecosystem. If HAV technology proves equally advantageous in the U.S. context – a reasonable expectation given global trends – similar outcomes are likely.

Civil engineering scholars point to the Netherlands’ “bottleneck memoranda” and subsequent implementation of “woonerfs” as globally applicable models of agile municipal policy (Bruno et al., 2021). While civil design ought to reflect local conditions, detailing specific road standards for every U.S. municipality is beyond this memo’s scope. Nonetheless, these two Dutch concepts offer instructive examples of constituent-based policy. The bottleneck memoranda, detailed in “Environmental Innovation and Societal Transitions”, is “a tool that relied on community participation in reporting obstacles to cycling”(Bruno et al., 2021). Like this request for public comment, effective civil design depends on an active civic environment (Smith, 2009, pp. 166–167).

As Frank Fischer and John Forester argue in “The Argumentative Turn in Policy Analysis and Planning”: “If analysts’ ways of representing policy and planning issues must make

assumptions about causality and responsibility, about legitimacy and authority, and about interests, needs, values, preferences, and obligations, then the language of policy and planning analyses not only depicts but also constructs the issues” (Fischer & Forester, 1993).

Evidence of this stagnation can be seen in U.S. civil infrastructure developments, where consistent adherence to interstate highways shapes discussion more than it permits alternatives. Currently, federal regulations on road structure inhibit local participation by imposing uniform standards that may not meet local needs. Deregulating road-based provisions within the FMVSS would restore municipal control over civic design and could usher in an era of infrastructure feedback and local implementation – as seen in the Netherlands.

Woonerfs further exemplify what constituent-focused design can achieve. Private vehicles currently account for an extremely high number of accidental deaths in the United States (National Center for Statistics and Analysis, 2025). The Netherlands, however, has mitigated this issue through the implementation of woonerfs – “low-speed traffic environments discouraging through-traffic and eliminating distinctions between pedestrians and car spaces” (Bruno et al., 2021). In practice, these spaces integrate vehicles, pedestrians, and cyclists. Accordingly, road safety, in a sense, monitors itself through peaceful coexistence rather than strict vehicle-based standards.

The distinct conditions that HAVs require have precipitated this request for public comment, demonstrating that the vestigial assumption of uniform private-vehicle applicability is no longer valid. While the current car-centric standard may appear to provide predictable scalability for HAV manufacturers (Rudnick, 2024), inevitable adherence to existing roads stifles industry-specific advancement. Patchwork infrastructure would instead create a varied and targeted environment for HAV implementation, benefiting both self-interested merchants and

consumers. When municipalities have the flexibility to design for their own contexts, community feedback can enable precisely the kind of innovation seen in the Netherlands (Bruno et al., 2021).

The provisions outlined in S14.9.3.12.5 are not exhaustively representative of road-based standards within FMVSS statutes. This comment, and any effective advocacy toward establishing a tech-neutral performance standard, must first address the vestigial laws that inhibit future innovation. Provisionally, the NHTSA may instead establish a tech-neutral standard in its own right to ensure road safety during this transitional period. It is for these reasons that the NHTSA ought to remove all regulations that include road-based standards: any reasoned effort toward encouraging innovation must first pivot from personal-vehicle-centric planning toward HAV-friendly prosperity.

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